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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/056,220	08/06/2018	Timothy R. Machold	MVRX1100-1	1072

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EXAMINER
ORKIN, ALEXANDER J

ART UNIT	PAPER NUMBER
3771	

NOTIFICATION DATE	DELIVERY MODE
07/07/2020	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

gtdocket@dlapiper.com

Office Action Summary	Application No. 16/056,220	Applicant(s) Machold et al.	
	Examiner ALEXANDER J ORKIN	Art Unit 3771	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 06/16/2020.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 40-77 is/are pending in the application.

5a) Of the above claim(s) 44 and 67-77 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 40-43 and 45-66 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 08/06/2018 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☑ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

2. Applicant's election without traverse of Invention I, claims 40-66 in the reply filed on 06/16/2020 is acknowledged.

3. Applicant's election of Species F; figure 21, BB; figure 8a,b claims 40-66 in the reply filed on 06/16/2020 is acknowledged. Because applicant did not distinctly and specifically point out the supposed errors in the restriction requirement, the election has been treated as an election without traverse (MPEP § 818.01(a)).

4. However, it is to be noted claim 44 is directed to subject matter of the posterior anchor is disposed along or adjacent the distal magnetic head, which is not supported by elected species F, BB, figures 21, 8a,b, but by figure 20 Species E. Therefore, claim 44 will be examined as not a part of the elected invention.

5. Claims 44, 67-77 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention and species, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 06/16/2020.

Priority

6. Acknowledgment is made of applicant's claim for foreign priority based on an application filed in IB on 02/07/2018. It is noted, however, that applicant has not filed a certified copy of the IB2018017175 application as required by 37 CFR 1.55.

Drawings

7. Color photographs and color drawings are not accepted in utility applications unless a petition filed under 37 CFR 1.84(a)(2) is granted. Any such petition must be accompanied by the appropriate fee set forth in 37 CFR 1.17(h), one set of color drawings or color photographs, as appropriate, if submitted via EFS-Web or three sets of color drawings or color photographs, as appropriate, if not submitted via EFS-Web, and, unless already present, an amendment to include the following language as the first paragraph of the brief description of the drawings section of the specification:

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

Color photographs will be accepted if the conditions for accepting color drawings and black and white photographs have been satisfied. See 37 CFR 1.84(b)(2).

8. The drawings are further objected to because figure 21 has a dotted line going off the page with no reference number.

9. The drawings are objected to under 37 CFR 1.83(a). The drawings must show every feature of the invention specified in the claims. Therefore, the “tether” of claim 66 must be shown or the feature canceled from the claims. No new matter should be entered.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure

number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Objections

10. Claim 45 is objected to because of the following informalities: Claim 45 line 2 should read "elongate member ~~having~~ includes a strain relief portion" or "elongate member having ~~includes~~ a strain relief portion". Appropriate correction is required.

11. Claim 65 is objected to because of the following informalities Claim 65 line 3 can read "~~ancho~~ anchor".

Claim Rejections - 35 USC § 112

12. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

> The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

13. Claims 40-43, 45-66 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and

distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

14. Claim 40 recites the limitation "the distal magnetic head" in line 10. There is insufficient antecedent basis for this limitation in the claim. Claim 40 recites “a magnetic head along a distal portion” but not “distal magnetic head”. Of note, the term “distal magnetic head is used in dependent claims 48, 49, 50, 53, and 59.

15. Regarding claim 43, the phrase "and/or" renders the claim indefinite because it is unclear whether the limitations preceding and following the phrase are both part of the claimed invention or separately part of the claimed invention.

16. Claim 49 is considered indefinite because it is unclear how the bridging element can extend from the posterior anchor and through the side hole and guide channel, and extend to a proximal end of the first catheter where it attached to a proximal end of the guidewire, *when* the guidewire is disposed within or proximal of the side whole during intravascular delivery. It is unclear the time point being claimed during “intravascular delivery” and when the bridge element is extended through the guide hole and connected to the guidewire.

17. Claim 65 recites the limitation "the posterior ancho(r) catheter" in line 10. There is insufficient antecedent basis for this limitation in the claim.

Claim Rejections - 35 USC § 102

18. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

19. Claims 40, 41, 45, 47-49, 53-64 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by U.S. Patent Publication 2008/0091264 to Machold.

As to claim 40, Machold discloses a delivery system for a heart implant for treatment of a heart valve in a heart chamber of a patient (paragraph 38), the system comprising: a first catheter (40) having a proximal and distal end, wherein the first catheter includes a first lumen (lumen as seen in figure 45a, 202 figure 45b) extending therethrough for passage of a guidewire (54, figure 45a,b), wherein the first catheter further includes: a magnetic head (200) along a distal portion thereof, wherein the magnetic head comprises magnetic poles oriented laterally relative a longitudinal axis of the first catheter (N, S, figure 45a,b) and a guide channel (where 105 extends, figure 45a,b) defined therein extending to a side hole (side hole as seen in figure 45a,b) adjacent a first magnetic pole, a penetrating guidewire (105) advanceable through a second lumen (figure 45b, paragraph 223, 225) aligned with the guide channel of the distal magnetic head, wherein the penetrating guidewire has a sharpened distal end to facilitate penetration of tissue (paragraph 223, 225); a posterior anchor (14) releasably coupled with the first catheter along the distal portion thereof (paragraph 217-219, 225, as the delivery catheter is delivered with respect to the system, it be releasably coupled); and a bridging element (12) attached to the posterior anchor at one end and attached to the penetrating guidewire at an opposite end such that advancement of the penetrating guidewire through the side hole and across the heart chamber facilitates advancement of the bridging element across the heart chamber (paragraph 194, 217-219, similar as seen in figure 27-28).

As to claim 41, Machold discloses the posterior anchor is axially offset from the magnetic head along the longitudinal axis of the first catheter and non-overlapping with the magnetic head (paragraph 225, as the anchor is delivered after the placement of the penetrating guidewire, the anchor will be axially offset.

As to claim 45, Machold discloses the posterior anchor is an elongate member having includes strain relief portion on each oppose end (122, figure 43a, paragraph 191, the flexible tube can be a strain relief).

As to claim 47, Machold discloses the posterior anchor comprises a center portion (132) that is substantially rigid and incudes an attachment feature (121) that securely attached to the bridging element (paragraph 191).

As to claim 48, Machold discloses the distal magnetic head includes a contoured recessed portion on one side dimensioned to accommodate the first lumen through the guide wires extends (figure 45a,b).

As to claim 49, Machold discloses the bridging element extends from the posterior anchor and through the side hole and guide channel of the distal magnetic head and extends to a proximal end of the first catheter where the bridge element attaches to a proximal end of the penetrating guidewire when the sharpened distal end of the penetrating guidewire is disposed within or proximal of the sidehole during intravascular delivery (figure 34d).

As to claim 53, Machold discloses the distal magnetic head of the first catheter comprises any of: a single magnet, a two magnet configuration, and a three magnet configuration (figure 45a,b).

As to claim 54, Machold discloses a second catheter (60) having a proximal and distal end, wherein the second catheter includes a first lumen (65) extending therethrough for passage of a guidewire, wherein the second catheter further includes: a magnetic head (210) along a distal portion thereof, wherein distal magnetic head comprises magnetic poles oriented axially relative a longitudinal axis of the second catheter and an axial channel defined therein extending to distal hole adjacent a first magnetic pole (figure 45a,b).

As to claim 55, Machold discloses the distal magnetic head of the second catheter is disposed at a distal most location on the second catheter (figure 45a-c).

As to claim 56, Machold discloses the distal magnetic head is configured such that the first magnetic pole of the first catheter is an opposite polarity as the first magnetic hole of the second catheter so as to facilitate a transverse alignment between the first and second catheters when magnetically coupled (figure 45b, paragraph 225).

As to claim 57, Machold discloses the first catheter includes one or more radio-opaque markers (the marker 1050, paragraph 281 usable with the embodiment of figure 45a-c) along a distal portion thereof that are asymmetrical about the longitudinal axis of the first catheter to facilitate alignment of rotational orientation of the first catheter relative the second catheter before magnetically coupling (paragraph 281).

As to claim 58, Machold discloses the one or more radio-opaque markers comprise first and second markers, the first marker being disposed on a same

side of the first catheter as the side hole and being different from the second marker under fluoroscopy (76 paragraph 175, and 1050, paragraph 181 usable with the embodiment of figure 45a-c).

As to claim 59, Machold discloses the distal magnetic heads of the first and second catheter are configured to remain magnetically coupled during advancement of the bridging element coupled to the posterior anchor through a penetrated tissue wall and into the second catheter (figure 45a-c). The catheter are *configured* to remain magnetically coupled.

As to claim 60, Machold discloses the first and second catheters are configured to remain substantially stationary relative to each other during deployment of the posterior anchor and advancement of the bridging element from the first catheter through the penetrated tissue wall and into the second catheter (figure 45a, paragraph 225).

As to claim 61, Machold discloses the system is configured such that advancement of the guidewire draws the bridging element attached to the posterior anchor across the chamber of the heart while the bridging element remains covered by the magnetically coupled catheters (figure 45a-c, paragraph 225, the system is *configured* for the positioning of the of bridging element as claimed).

As to claim 62, Machold discloses the system is configured such that deployment of the posterior anchor from the first catheter and advancement of the bridging element across the heart are performed by a single operator (paragraph 225).

As to claim 63, Machold discloses the penetrating guidewire has a length greater than the total length of the first and second lumens of the first and second catheters so as to allow manual advancement of the penetrating guidewire externally from a first vascular access point associated with the first catheter until the penetrating guidewire extends through the second lumen of the second catheter externally of a second vascular access point associated with the second catheter (paragraph 213, figure 45a-c).

As to claim 64, Machold discloses the bridging element has a length greater than a length of the penetrating guidewire so that at least a portion remains outside of a body of the patient as the bridging element is drawn through the first and second catheters (paragraph 214).

Claim Rejections - 35 USC § 103

20. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

21. The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

22. Claim 46 is rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication 2008/0091264 to Machold in view of U.S. Patent Publication 2008/0103527 to Martin et al

As to claim 46, Machold discloses the device above but is silent about the helical cut portion.

Martin teaches a similar device having a similar posterior anchor as Machold (flexible sheath paragraph 13 of Martin, "flexible anchor" as in Machold paragraph 191) with a helical cut portion for the purpose of using a similar tube mechanism for attaching to a bridge element (paragraph 13). The cut tube of Martin can be used as the anchor of Machold without any unexpected results. It would have been obvious to one of ordinary skill in the art before the effective filing date of the invention to use the anchor with the cut portion of Martin with the device of Machold in order for using a similar tube mechanism for attaching to a bridge element.

23. Claim 65 is rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication 2008/0091264 to Machold in view of U.S. Patent Publication 2005/0119734 to Spence.

As to claim 65, Machold disclose the device above but is silent about the smooth reinforced guidewire lumen that inhibits kinking during advancement of the catheter.

Spence teaches a similar device having smooth reinforced guidewire lumen that inhibits kinking during advancement of the catheter (paragraph 144) for the purpose of providing support to catheter as it is deflected. It would have been obvious to one of ordinary skill in the art before the effective filing date of the invention to use the smooth reinforced guidewire lumen of Spence with the device of Machold in order for providing support to catheter as it is deflected

Conclusion

24. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. U.S. 6,190,353 to Makower, U.S. Patent 6,352,543 to Cole, U.S. Patent 6,464,665 to Hueser and U.S. Patent Publication 2015/0250470 to Vargas all disclose similar device readable or combinable on the claims of record.

25. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALEXANDER J ORKIN whose telephone number is (571)270-7412. The examiner can normally be reached on Monday - Friday 9am - 5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Elizabeth Houston can be reached on (571)272-7134. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppair-my.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALEXANDER J ORKIN/
Primary Examiner, Art Unit 3771